Boyer Moore

```
#include <iostream>

#include <string>

using namespace std;

#define NO_OF_CHARS 256

void badCharHeuristic(string str, int size, int badchar[NO_OF_CHARS]) {

	for (int i = 0; i < NO_OF_CHARS; i++)

		badchar[i] = -1;

	for (int i = 0; i < size; i++)

		badchar[(int)str[i]] = i;

}

void search(string txt, string pat) {

	int m = pat.size();

	int n = txt.size();

	int badchar[NO_OF_CHARS];

	badCharHeuristic(pat, m, badchar);

int s = 0;

	while (s <= (n - m)) {

		int j = m - 1;

		while (j >= 0 && pat[j] == txt[s + j])

			j--;

		if (j < 0) {

			cout << "Pattern occurs at index " << s << endl;

			s += (s + m < n) ? m - badchar[txt[s + m]] : 1;

		} else {
```

```
                s += max(1, j - badchar[txt[s + j]]);
            }
        }
}
int main() {
        string txt = "ABAAABCD";
        string pat = "ABC";
        search(txt, pat);
        return 0;
}
```

Output

```
Pattern occurs at index 4


=== Code Execution Successful ===
```